SE311 SPRING 2025-2026 / 13-05-2026
Lab Work 10 – Visitor Pattern

Imagine you are the architect of a Hospital Management System that monitors three distinct patient groups: **`Newborn Infants`**, **`Post Surgery Adults`**, and **`Elderly Residents`**. Each group is connected to different monitoring equipment and has unique biological benchmarks for things like heart rate, mobility, and sleep patterns.

As the hospital expands its services, various departments need to process patient data for different reasons. The **`Billing Department`** must calculate daily costs based on the specific equipment used by each group. Simultaneously, the **`Research Division`** wants to gather anonymous statistics on recovery trends across all ages. (You can simply print some statements for that part in the related methods.)

Instead of coding financial math or statistical analysis directly into the patient records, you as a architect create a system where the records stay clean and focused only on medical data. To manage these patients efficiently, you implement a **`Hospital Management System`** as a central structure that maintains a comprehensive list of all active patient records. You establish a standard way for authorized service teams to interact with each group by passing these teams through the Hospital Management System's structure.

This system allows the hospital to launch an infinite number of new administrative or medical relatives by simply introducing a new specialized team to handle the patient's data in its own way without ever modifying the core Hospital Management System's software.

Now, think about "If we introduce a new type of patient like a **`'Critical Care Patient'`** what changes do we need to make to our current structure?"